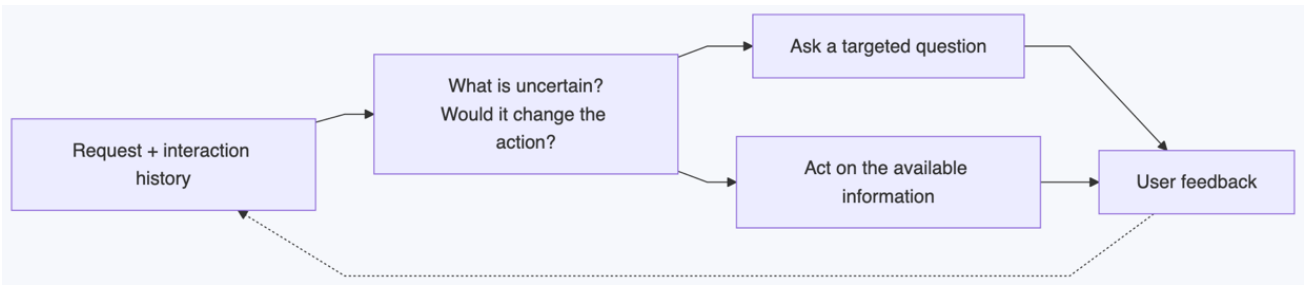
On Long-Running Agent for User’ s ImplicIt Intention Inference (I3Agent)

I3Agent Project Online Update Page (Notion): **https://www.notion.so/3ff200d7109280699628d1cf59d08625**

A. A Narrow Goal

Build a proactive agentic personal AI assistant that knows when to ask and what to ask. (methodology contribution)

Working hypothesis. Asking when an agent is uncertain could change the next action and improve task success over confidence-based questioning, at the same question and user burden.

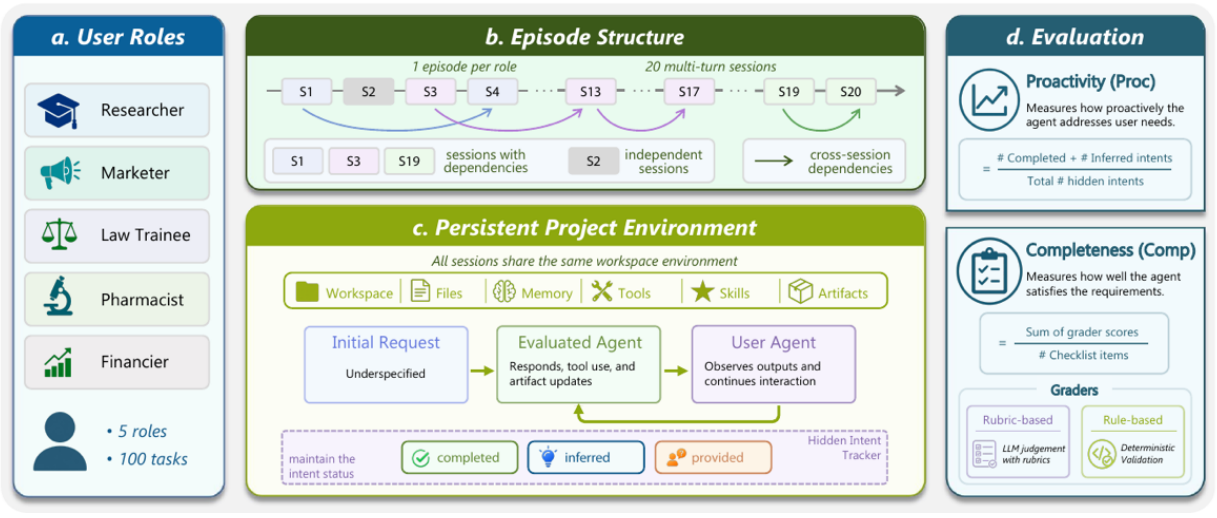


B. Related Source

Environments and adaptation: **π-Bench** tests proactive tool use with hidden requirements; **LifeSim** simulates user intentions and preferences across conversations; **PAST-Bench** tests whether personal agents improve through retained experience.

Recent baseline candidates:

- **VoI** and **REVOIR**: inference-time decisions about whether clarification is worth its cost.
- **Steerable Clarification**: self-play training to choose between asking, guessing, and answering multiple interpretations under explicit costs.
- **Pareto-Dominant Clarification** (Oct. 2026): RL for coding accuracy under a question budget.
- **Learning to Clarify Underspecified Intents** (Oct. 2026): learning which questions best recover user utility in image generation.



*π-Bench overview (**original paper, Figure 1**).*

C. Pilot Study

Step	Smallest useful experiment
Reproduce	Run π-Bench tasks.
Inspect	Identify useful questions, unnecessary questions, and mistaken assumptions.
Compare	Retrieval-and-act, confidence-based questioning, and history-matched VoI. Test a targeted policy suggested by the analysis.

Share the model, history, tools, and question budget. Measure **success, wrong assumptions, questions, and inference cost**. Continue if targeted questioning improves success at matched cost across scripted and native simulated users.

D. Failure mode & Research Ideas

Main risks: exploiting the user simulator, or mistaking execution failures for intention failures. Check final outputs and rerun failures with the true requirement supplied.

Four ideas to explore:

1. **Self-improvement:** learn how to adapt through context, memory, or weight updates.
2. **Shared intentions:** build future-triggered plans with the user, including revisions and cancellation.
3. **Questioning as control:** optimize when and what to ask through decision-making or RL.
4. **Uncertainty:** represent both how uncertain the assistant is and what it is uncertain about.